Problem Set VII: Edgeworth Box, Robinson Crusoe



Paolo Crosetto
paolo.crosetto@unimi.it

DEAS - University of Milan

Exercises will be solved in class on *March 22nd, 2010*



Extra CLASSES



Extra classes

Two extra classes are scheduled, one for discussing the solution to Problem Set Marked 3, the other devoted to a Question & Answer session.

1. Aula 27, Monday 29th of March, 14.30 - 16.15 - Solution to PSet Mark3
2. Aula 21, Tuesday 30th of March, 15.00 - 17.00 - Question & Answers (all program)

Problem Set MARKED 3



It's Long!

Keep in mind that the problem set is longer than previous ones. But...

- It is useful for a revision of most material of the course
- if you do it well, the exam should be a breeze.

Good luck!

1. Edgeworth box

Consider a pure-exchange, private-ownership economy, consisting in two consumers, denoted by $i = 1, 2$, who trade two commodities, denoted by $l = 1, 2$. Each consumer i is characterized by an endowment vector, $\omega_i \in \mathbb{R}^2_+$, a consumption set, $X_i = \mathbb{R}^2_+$, and regular and continuous preferences, $\succsim_i$ on X_i. Initial endowments are given by $\omega_1 = (4, 2)$ and $\omega_2 = (2, 3)$. Individual utility functions are $u_1(x_{11}, x_{21}) = x_{11}x_{21}$ and $u_2(x_{12}, x_{22}) = x_{12} + x_{22}$.

1. Draw the Edgeworth Box for this economy, drawing endowment point ω and the indifference curves passing through it for both consumers.
2. Find analytically the Pareto Set (interior points and, separately, boundary points) and the Contract curve. Draw them both in the Edgeworth Box.
3. Find the competitive equilibrium prices and allocations. Draw it in the Edgeworth Box.
4. Take the point $\tilde{x} = ((\tilde{x}_{11}, \tilde{x}_{21}), (\tilde{x}_{12}, \tilde{x}_{22})) = ((1, 1), (5, 4))$. Draw it in the Edgeworth box, show that it is in the Pareto set, and write down the implicit shadow prices. Compute the transfers T_1 and T_2, with $T_1 + T_2 = 0$, that ought to be assigned to either consumer in order that, starting from the initial endowments, the allocation concerned can be obtained as a price equilibrium with transfers.

Setting up the box



- The dimension of the box are given by $\bar{\omega}_1 = 4 + 2 = 6$ and $\bar{\omega}_2 = 2 + 3 = 5$;
- The indifference curves are level sets of the utility functions:

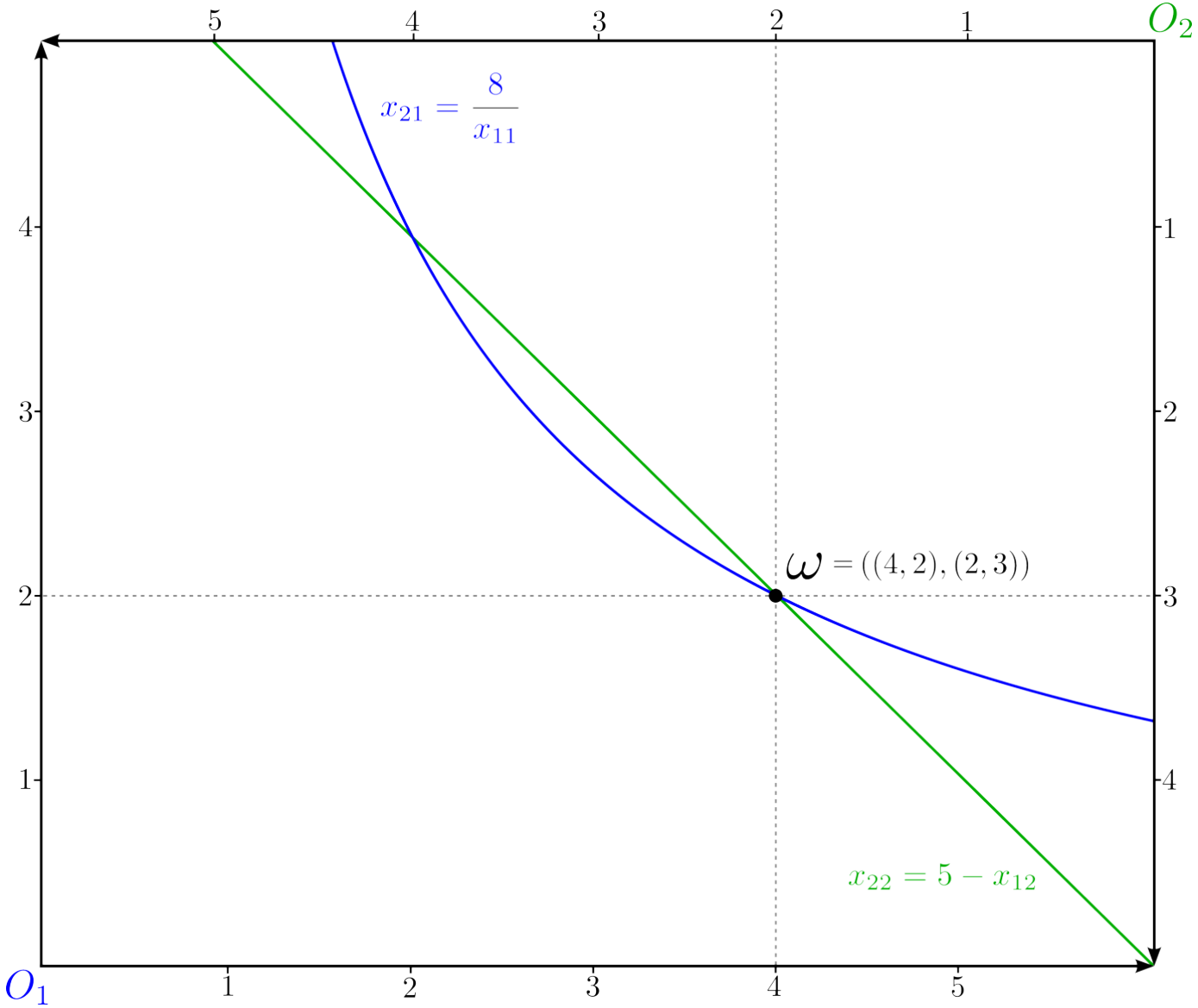
$$\text{For 1: } x_{11}x_{21} = u_1(\omega) \qquad \text{For 2: } x_{12} + x_{22} = u_2(\omega)$$

- The utility level of endowment is $u_1(\omega) = 4 \cdot 2 = 8$ for consumer one and $u_2(\omega) = 2 + 3 = 5$ for consumer two;
- Hence, the indifference curves passing through the endowment point ω are:

$$\text{For 1: } x_{21} = \frac{8}{x_{11}} \qquad \text{For 2: } x_{22} = 5 - x_{12}$$

- Note that consumer 1 has Cobb-Douglas preferences, while consumer 2 has perfect substitutes;
- See the drawing below for graphical representation.

Edgeworth Box and indifference curves, graphics



Pareto Set, interior points



- The interior points of the Pareto Set are given by all the points in which the two MRS_{12}^{i} are equal.

$$MRS_{12}^{1} = \frac{\partial u_1}{\partial x_{11}} \left(\frac{\partial u_1}{\partial x_{21}} \right)^{-1} = \frac{x_{21}}{x_{11}}$$

$$MRS_{12}^{2} = \frac{\partial u_2}{\partial x_{12}} \left(\frac{\partial u_2}{\partial x_{22}} \right)^{-1} = \frac{1}{1} = 1$$

- Hence, the interior points for player one are given by the condition

$$x_{21} = x_{11} \text{ w.r.t. } O_1$$

- and, for player two, by

$$5 - x_{22} = 6 - x_{12} \quad \Rightarrow \quad x_{22} = x_{12} - 1 \text{ w.r.t. } O_2$$

Pareto Set, boundary points, Contract Curve



- There are boundary points belonging to the Pareto Set: the Pareto Set always starts from the two origins of the EB.
- Hence all points for which $x_{22} = 0$ and $x_{12} \leq 1$ are in the Pareto Set.
- This is because at those points, consumer 2 would like to sell more of good 2, but he can't, while consumer 1 is willing to buy more of both goods, but he can buy more only of good 1 since he holds already all of good 2.
- See the graphics for intuition.

- The contract curve is all the points in the Pareto Set for which utility is higher or equal than the utility given by the endowments.
- Hence, its lower-leftmost point is given by the intersection of the pareto set with the endowment utility of player 1:

$$\begin{cases} x_{11}x_{21} = 8 \\ x_{21} = x_{11} \end{cases} \quad \Rightarrow \quad x_{11}^2 = x_{21}^2 = 8 \quad \Rightarrow \quad x_{11} = x_{21} = 2\sqrt{2}$$

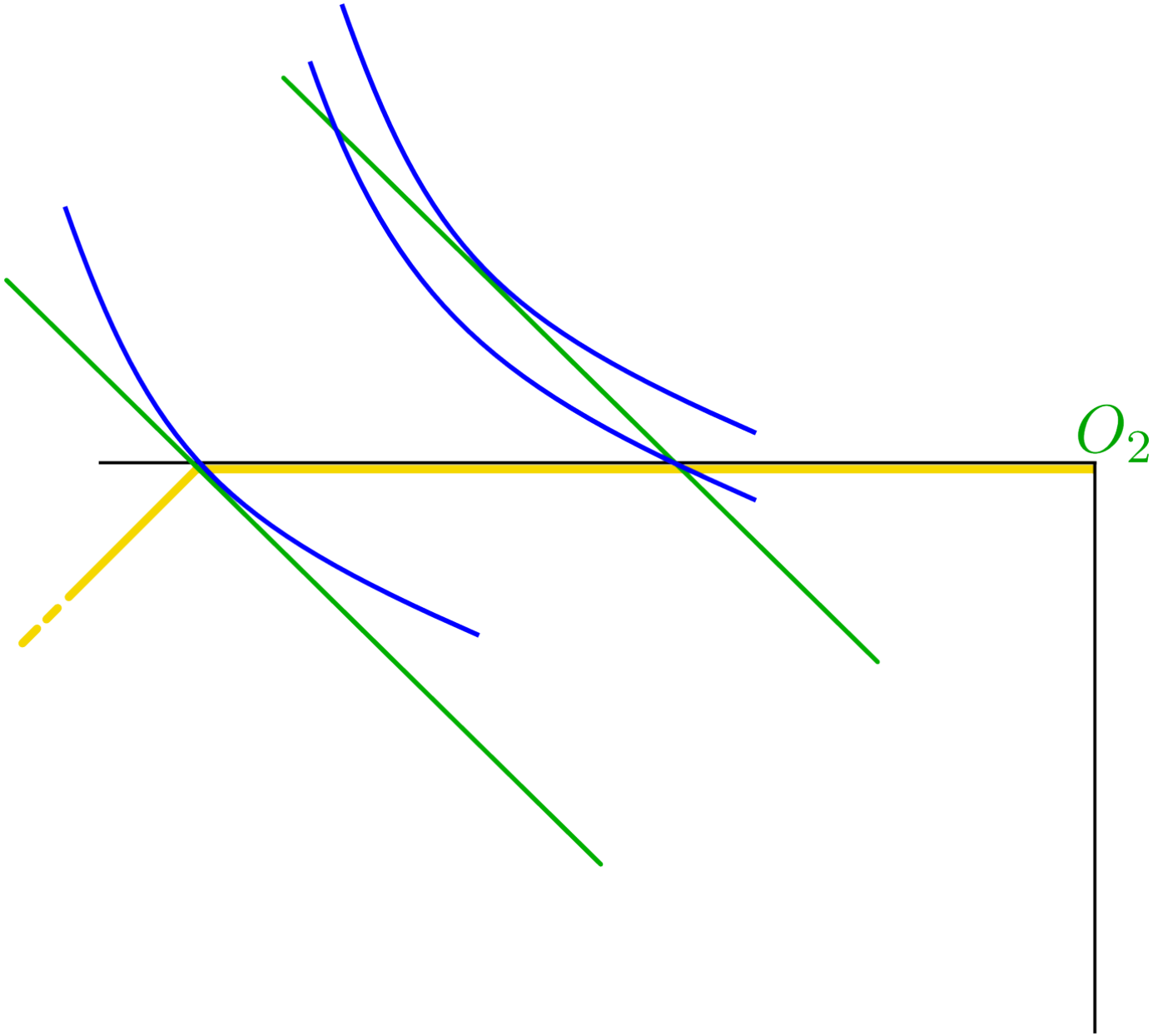
- Similarly, the upper-rightmost point for player 2 is given by:

$$\begin{cases} x_{12} + x_{22} = 5 \\ x_{22} = x_{12} - 1 \end{cases} \quad \Rightarrow \quad 2x_{12} = 6 \quad \Rightarrow \quad x_{12} = 3; \quad x_{22} = 3 - 1 = 2$$

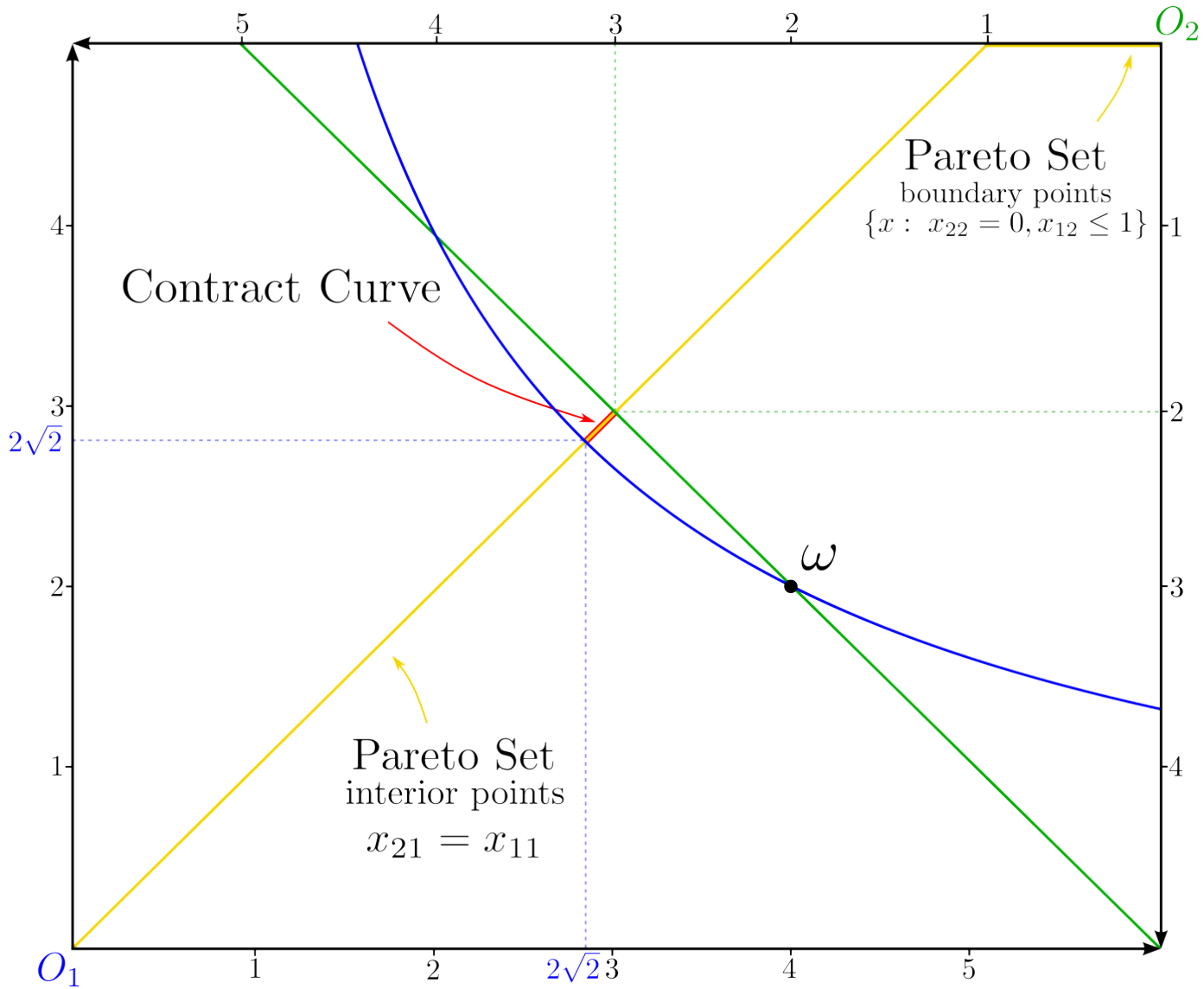
- The Contract curve, making reference to the origin of consumer one, is:

$$\{x \in \text{Pareto Set} : \ 2\sqrt{2} \leq x_{11} \leq 3 \,\text{and}\, 2\sqrt{2} \leq x_{21} \leq 3\}$$

Pareto Set, boundary points, graphics



Pareto Set, Contract Curve, graphics



Equilibrium



- In equilibrium, both players optimize given prices, and prices move to guarantee market clearing.
- Exploiting *MRS* calculations, for consumer one, imposing $p_1 \equiv 1$:

$$MRS_{12}^1 = \frac{p_1}{p_2} \quad \Rightarrow \quad \frac{x_{21}}{x_{11}} = \frac{1}{p_2}$$

$$MRS_{12}^2 = \frac{p_1}{p_2} \quad \Rightarrow \quad 1 = \frac{1}{p_2} \quad \Rightarrow \quad p_2^* = 1$$

- Plugging the equilibrium price system $p^* = (1, 1)$ into the budget constraints we get demand functions for player 1 and 2.
- For consumer 1:

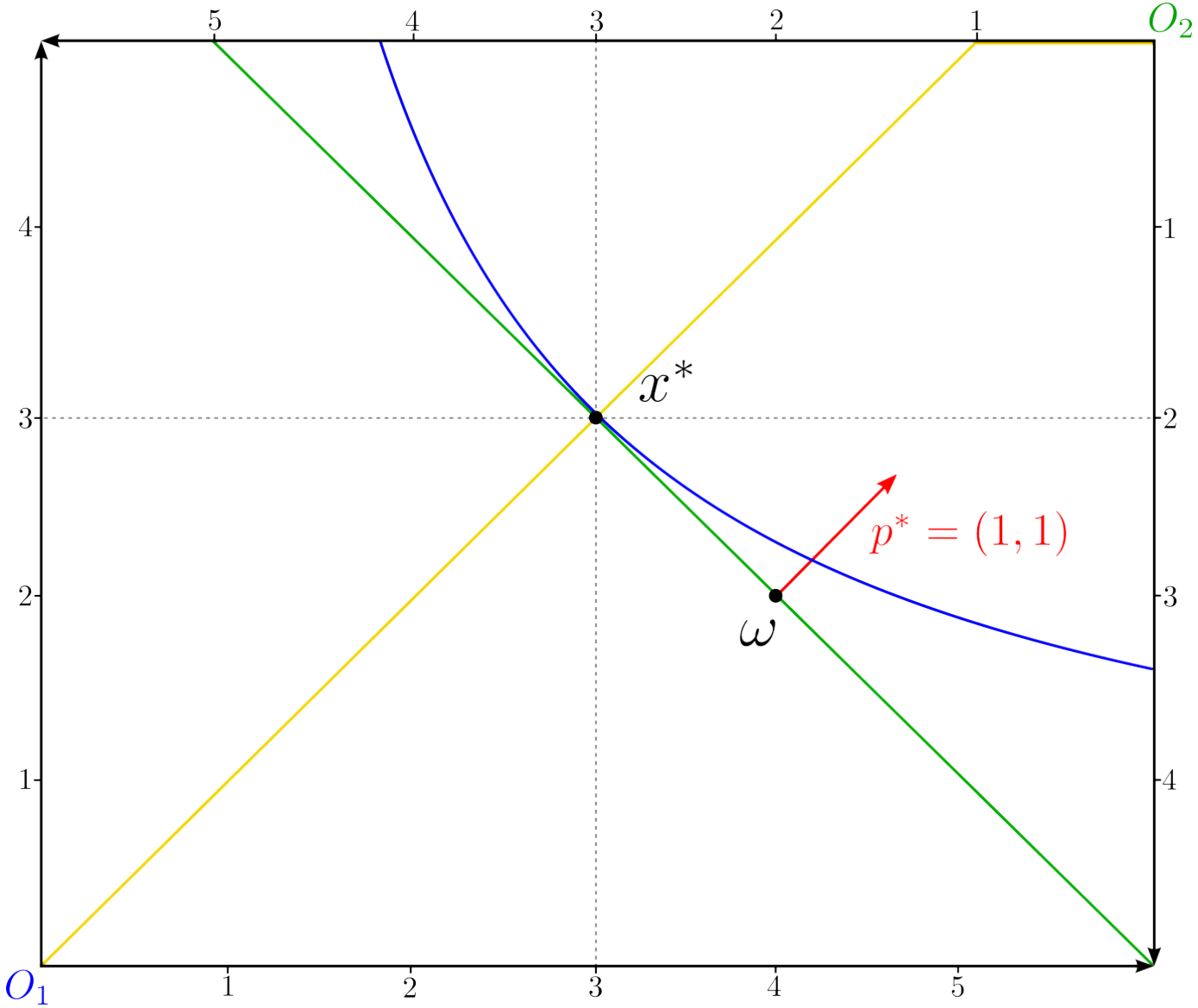
$$\begin{cases} x_{11} = x_{21} \\ x_{11} + x_{21} = 6 \end{cases} \quad \Rightarrow \quad x_{11}^* = x_{21}^* = 3$$

- and, for consumer 2:

$$\begin{cases} x_{22} = x_{12} - 1 \\ x_{12} + x_{22} = 5 \end{cases} \quad \Rightarrow \quad x_{12}^* = 3;\ x_{22}^* = 2$$

- Hence, equilibrium is $x^* = ((3, 3), (3, 2))$, and $p^* = (1, 1)$.

Edgeworth Box equilibrium, graphics



Price equilibrium with transfers



- An allocation in the Edgeworth Box can be sustained as a *Price Equilibrium with Transfer*;
- That is, a planner can make transfers and then let the consumers freely exchange, in order to lead them to a desired solution
- In this case the planner wants to support as an equilibrium the allocation $\tilde{x} = ((1,1),(5,4))$
- The planner must hence use transfers T_1, T_2 s.t.: $T_1 + T_2 = 0$

1. The point is in the Pareto Set: $\tilde{x}_1$ satisfies $x_{11} = x_{21}$ (and hence it satisfies the condition for the other consumer, too, no need to check).
2. The implicit shadow prices are constant in all of the Pareto Set, and were calculated to be $\tilde{p} = p^* = (1,1)$.
3. The Transfer must be calculated by comparing the value of endowment and the value of allocation $\tilde{x}$ for both consumers at prices $p^* = (1,1)$.

Calculating transfers



- A planner must transfer wealth in order to match for the difference in value at prices $p^* = (1, 1)$ between the endowments and the preferred point, in this case $\tilde{x} = ((1, 1), (5, 4))$

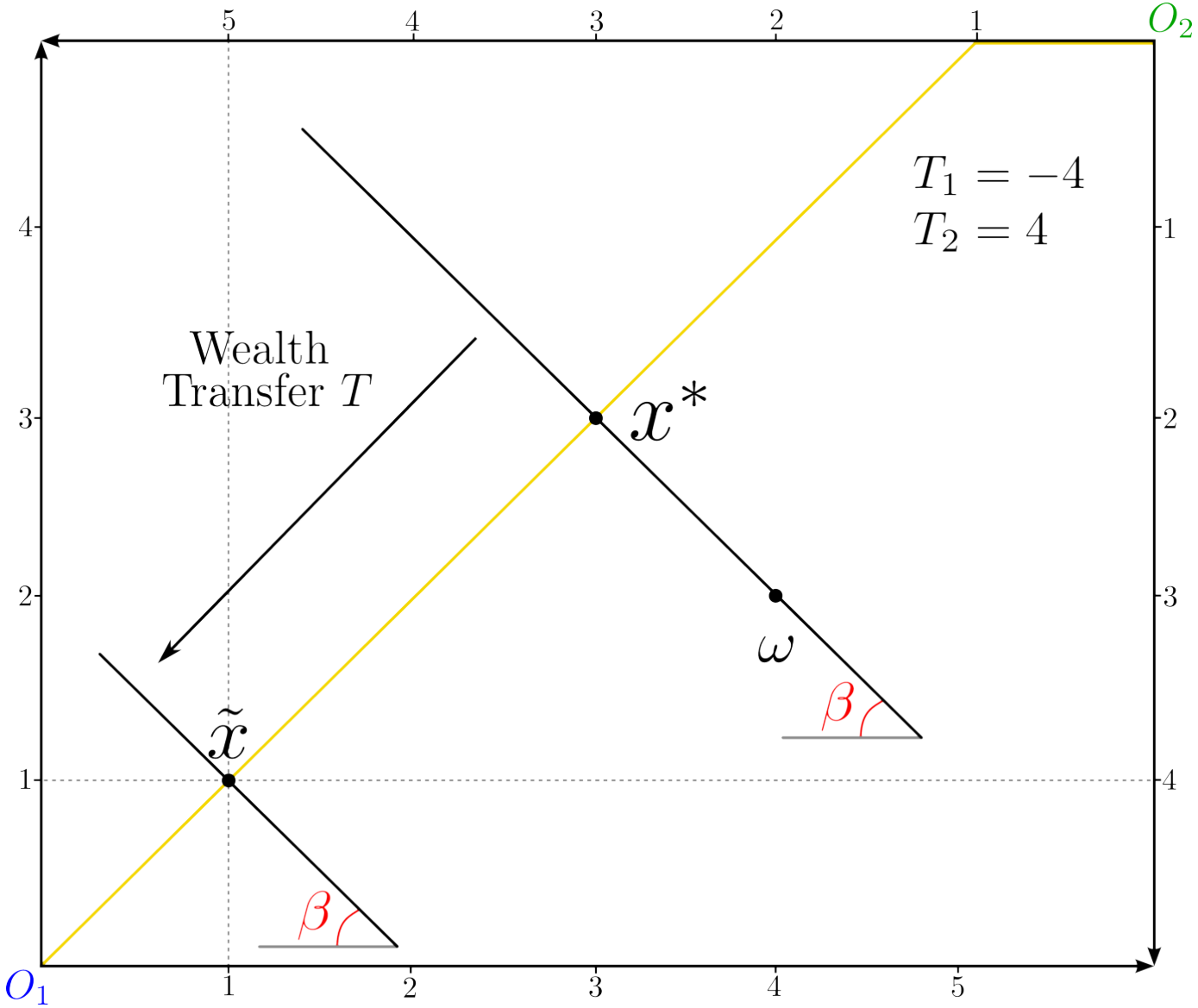
For 1: $p^* \cdot \tilde{x} = (1, 1) \cdot (1, 1) = 1 + 1 = 2 \quad p^* \cdot \omega_1 = (1, 1) \cdot (4, 2) = 4 + 2 = 6$

For 2: $p^* \cdot \tilde{x}_2 = (1, 1) \cdot (5, 4) = 5 + 4 = 9 \quad p^* \cdot \omega_2 = (1, 1) \cdot (2, 3) = 2 + 3 = 5$

$$T_1 = p^* \cdot \tilde{x}_1 - p^* \cdot \omega_1 = 2 - 6 = -4; \quad T_2 = p^* \cdot \tilde{x}_2 - p^* \cdot \omega_2 = 9 - 5 = 4;$$

$$T_1 = -4; \quad T_2 = 4; \quad T_1 + T_2 = 0.$$

Price equilibrium with transfers, graphics



Recap: Robinson Crusoe Economy



The simplest possible general equilibrium model including production is one with two economic agents, 1 consumer and 1 producer, superimposed into one person only: lonely Robinson on his remote island.

- Note that price-taking behaviour is assumed for both producer and consumer;
- note as well that the consumer owns the firm and earns all of its profits.
- These two assumptions do not make much sense in 1-person economies, but the model is just an extreme simplification.

Recap: Robinson's producer problem

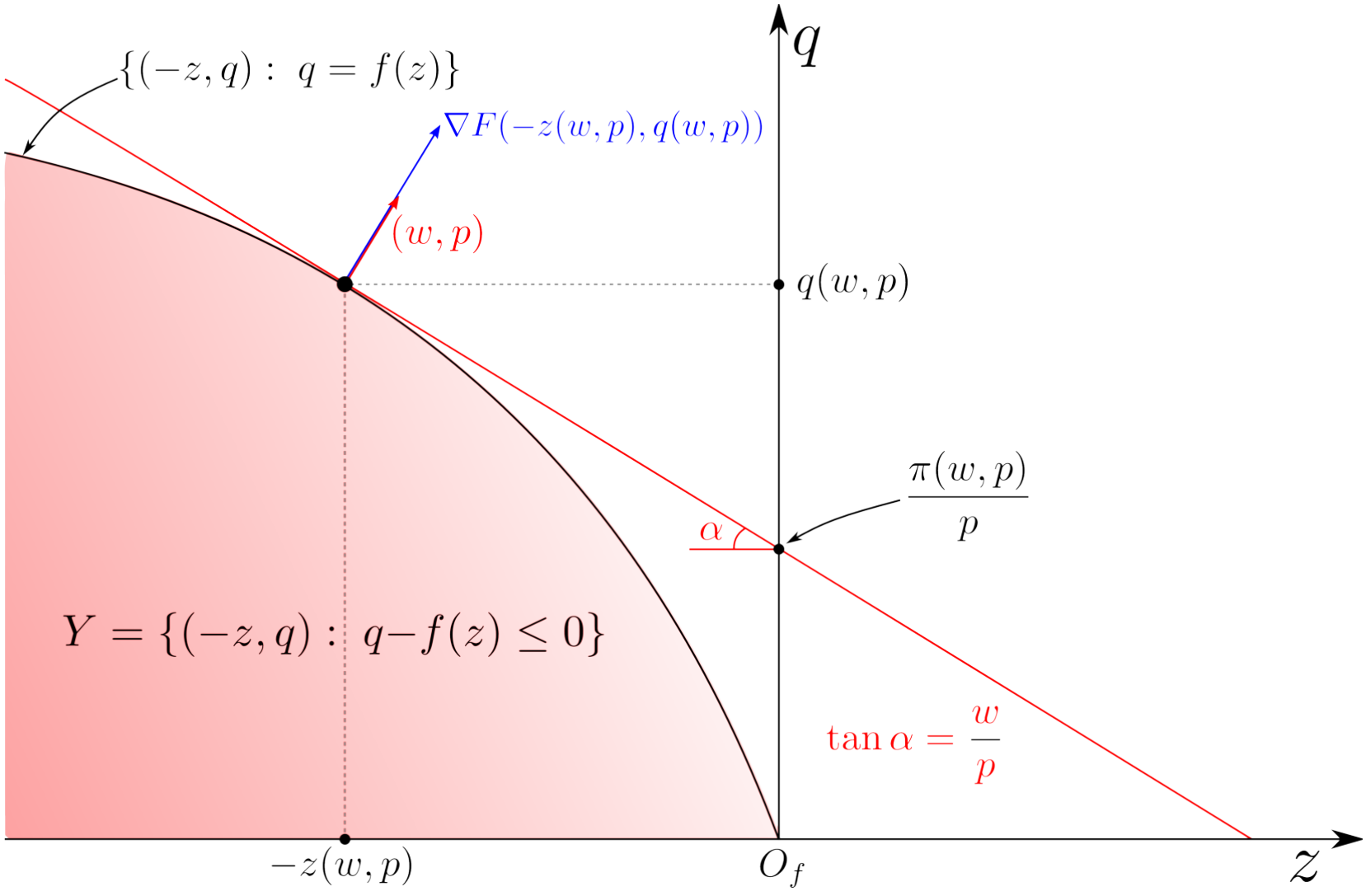


- Robinson as a producer acts as a competitive firm: maximises profits given technology.
- He hence solves

$$\max_{z \geq 0} \; pf(z) - wz$$

- And from this maximisation problem derives the firm's labor demand $z(w, p)$, output $f(z(w, p)) = q(w, p)$, and profits $\pi(w, p)$.

Robinson as a producer



Recap: Robinson's consumer problem

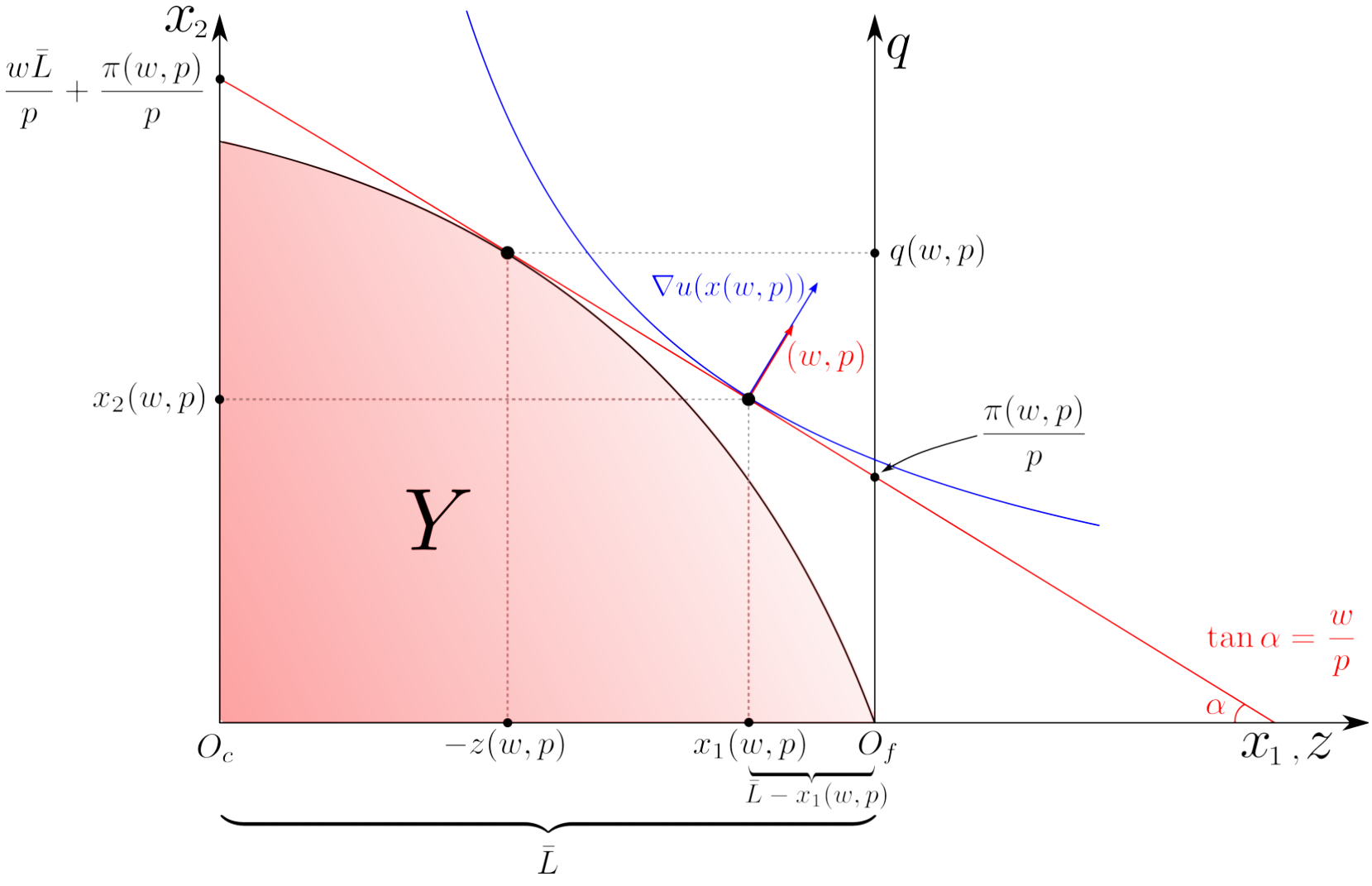


- The consumer owns the firm: he hence solves a utility maximisation on two goods, leisure and a consumer good, having as resources his time ($\bar{L}$) and the profits generated by the firm.
- He hence solves

$$\max \quad u(x_1, x_2) \quad \text{s.t. } wx_1 + px_2 = \bar{L} + \pi(w, p)$$

- From which the two walrasian demands $x_1(w, p)$ and $x_2(w, p)$ can be derived,
- jointly with the labour supply function, $s(w, p) = \bar{L} - x_1(w, p)$.

Robinson as a consumer



Recap: Robinsonian equilibrium

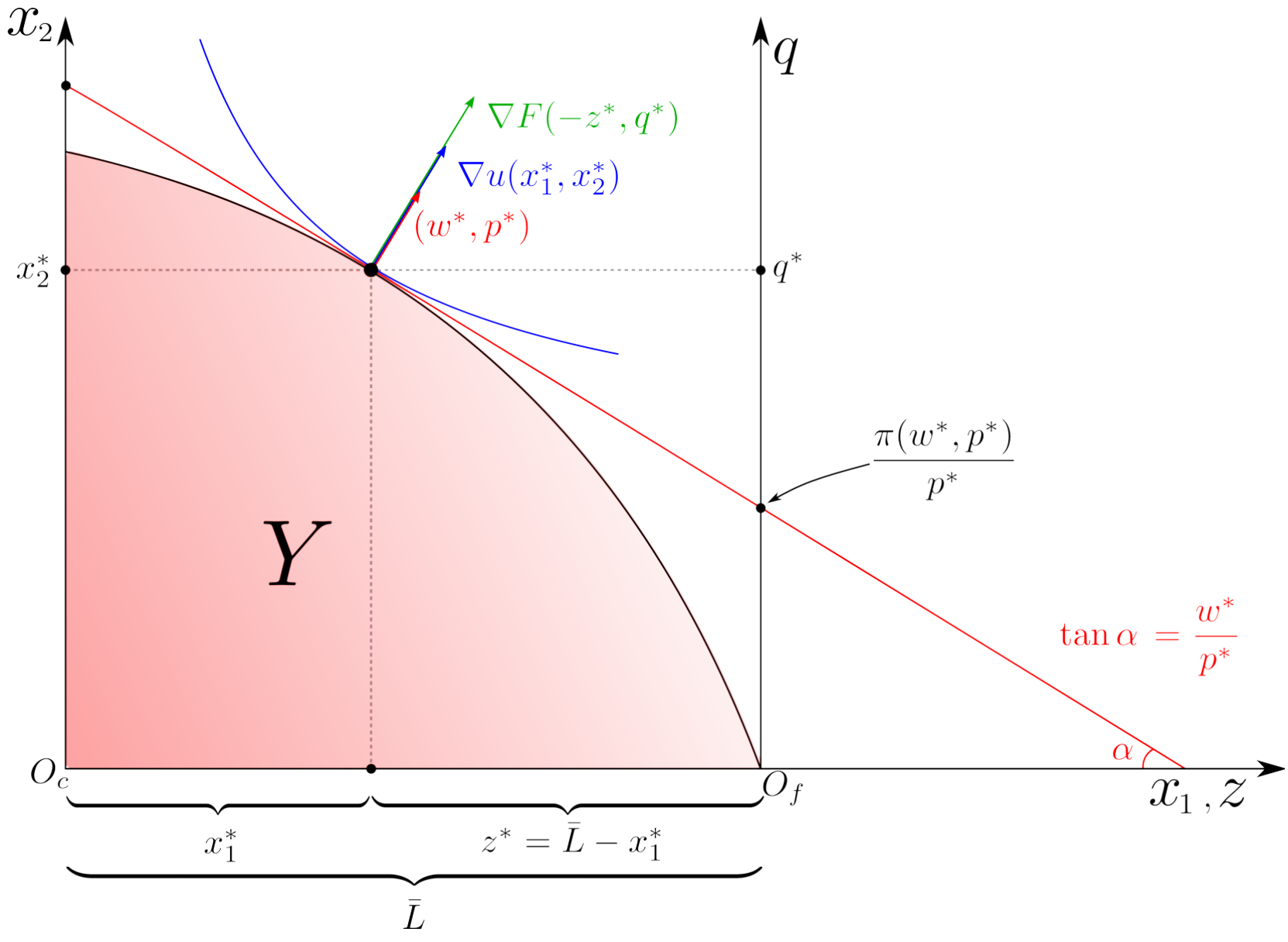


- An equilibrium is a price vector (w^*, p^*) at which both consumption and labour market clear,
- that is, at which

$$x_2(w^*, p^*) = q(w^*, p^*) \text{ and } z(w^*, p^*) = \bar{L} - x_1(w^*, p^*)$$

- An equilibrium maximises utility subject to the technology and endowment constraints
- In this case, if functions are well-behaved, both theorems of welfare economics hold.

Robinsonian equilibrium



Properties of production function



- $f(z)$ shows *constant returns to scale*:

$$f(\alpha z) = 2\alpha z = \alpha f(z)$$

- The marginal product of labour is constant:

$$\frac{\partial f(z)}{\partial z} = 2$$

- Average product of labour is constant:

$$\frac{f(z)}{z} = 2$$

- We know that in the case of *CRS* the profit maximisation is not well defined.
- We will have to determine the equilibrium on the consumer side.

Profit maximisation problem,I



Robinson as a producer solves a profit maximisation problem:

$$\max_{z \geq 0} \quad f(z) - wz$$

And the Lagrangean can be set up to be:

$$\max \quad L(z, \lambda) f(z) - wz + \lambda(-z)$$

And the FOC for this problem (Kuhn-Tucker), are:

$$\begin{cases} 2 - w = \lambda \\ \lambda \geq 0 \\ \lambda z = 0 \end{cases} \Rightarrow \begin{cases} 2 \leq w \\ (2 - w)z = 0 \end{cases}$$

That can be solved to yield

$$z(w) = \begin{cases} 0 & \text{for } w > 2 \\ \in [0, \infty) & \text{for } w = 2 \\ \infty & \text{for } w < 2 \end{cases}$$

Profit Maximisation Problem, II



The supply function $f(z(w))$ can then be calculated to be

$$q(w) = f(z(w)) = \begin{cases} 0 & \text{for } w > 2 \\ \in [0, \infty) & \text{for } w = 2 \\ \infty & \text{for } w < 2 \end{cases}$$

And the corresponding profit function can be calculated as usual by plugging the input demand correspondence into the maximand:

$$\pi(w) = \max_{z \geq 0} \{pf(z) - wz\} = \max_{z \geq 0} \{2z - wz\} = \begin{cases} 0 & \text{for } w \geq 2 \\ \infty & \text{for } w < 2 \end{cases}$$

Infinite profits imply an infinite demand for labour. This is not compatible with general equilibrium, since the supply of labour by Robinson-consumer cannot be infinte. Hence, the only value of profits that is consistent with a competitive equilibrium is $\pi(w) = 0$

Properties of utility function



This is a Cobb-Douglas utility function. It has the standard C-D properties:

$$u_1'(x_1, x_2) = \frac{\partial u(x_1, x_2)}{\partial x_1} = \frac{2}{3}\left(\frac{x_2}{x_1}\right)^{\frac{1}{3}}$$

$$u_2'(x_1, x_2) = \frac{\partial u(x_1, x_2)}{\partial x_2} = \frac{1}{3}\left(\frac{x_2}{x_1}\right)^{\frac{2}{3}}$$

$$MRS_1 2(x_1, x_2) = \frac{u_1'(x_1, x_2)}{u_2'(x_1, x_2)} = 2\frac{x_2}{x_1}$$

Hence, the gradient is always positive, $\nabla u(x_1, x_2) \gg 0$, for $x \in \mathbb{R}^2_+$.
Moreover, checking the bordered Hessian (do it!) we can conclude that the function is strictly quasi-concave.

Utility Maximisation problem, I



Robinson as a consumer solves a standard UMP, introducing in his budget constraint the maximal profits. In this case, $\pi(w) = 0$, so the problem, setting $p \equiv 1$, boils down to

$$\max_{x \geq 0} \; x_1^{\frac{2}{3}} x_2^{\frac{1}{3}} \quad \text{s.t.} \;\; wx_1 + px_2 = w\bar{L} + \pi(w) = 24w + 0 = 24w$$

FOCs boil down to

$$\begin{cases} MRS_{12}(x_1, x_2) = \dfrac{w}{p} \\ wx_1 + px_2 = w\bar{L} + \pi(w) \end{cases} \Rightarrow \begin{cases} 2\dfrac{x_2}{x_1} = w \\ x_2 = (24 - x_1)w \end{cases}$$

That can be easily solved to yield the demand for leisure $x_1(w)$ and the demand for consumer good $x_2(w)$:

$$x_1(w) = 16 \qquad x_2(w) = (24 - 16)w = 8w$$

The corresponding labour supply function, $s(w)$, can then be easily calculated:

$$s(w) = \bar{L} - x_1(w) = 24 - 16 = 8$$

Equilibrium



Equilibrium in the labour market implies:

$$s(w^*) = z(w^*) \quad \Rightarrow \quad z(w^*) = 8$$

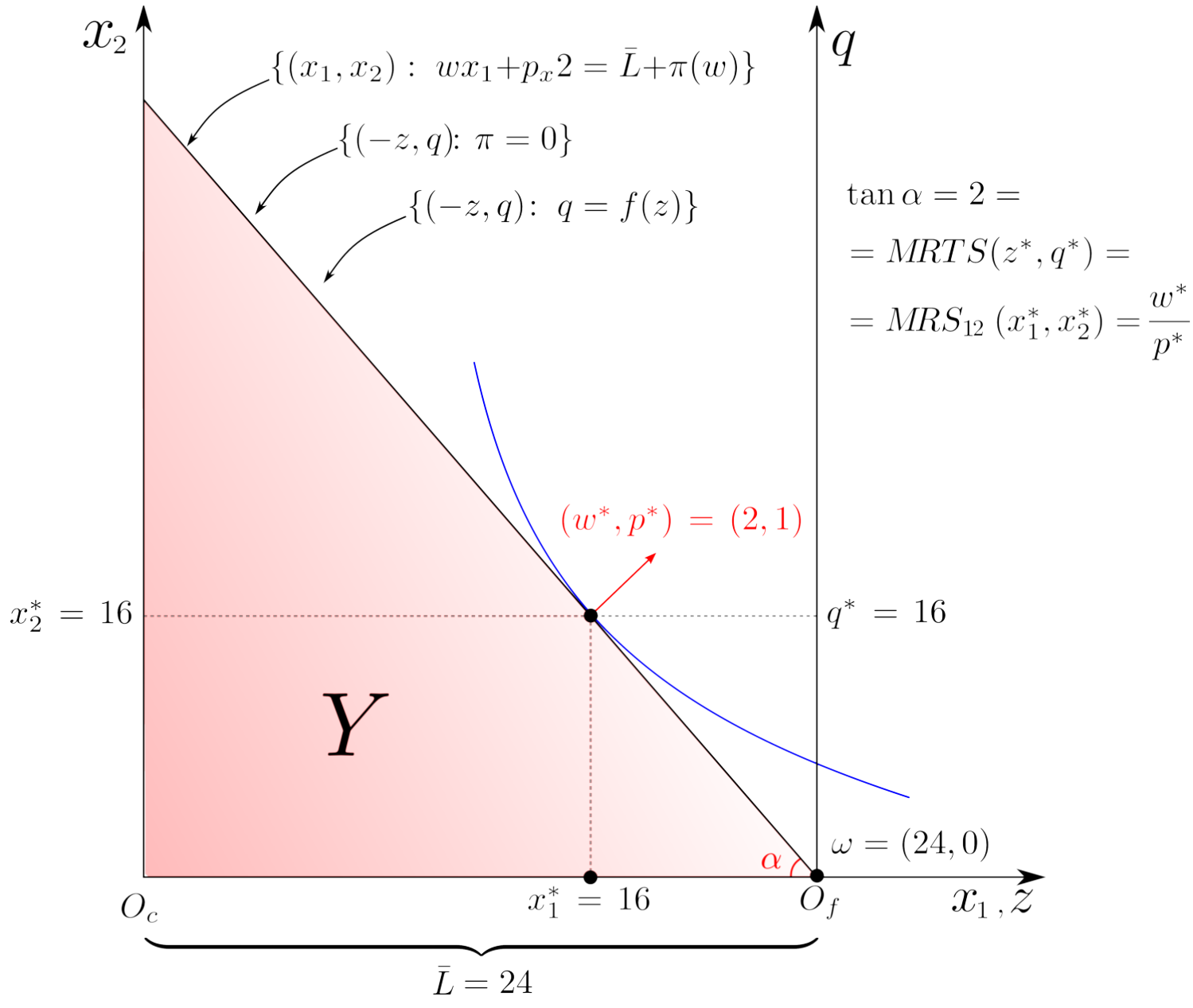
And equilibrium on the consumer good market in turn implies:

$$x_2(w^*) = q(w^*) \quad \Rightarrow \quad x_2(w^*) = 2(z(w^*)) = 16 = 8w^* \quad \Rightarrow \quad w^* = 2$$

Summing up, the equilibrium allocation and prices are given by:

$$E = (((x_1^*, x_2^*), (-z^*, q^*)), (w^*, p^*)) = (((16, 16), (-8, 16)), (2, 1))$$

Robinson Crusoe: graphics



The Walrasian equilibrium is the only Pareto Efficient allocation



Proof.

- In this case both the First and the Second Fundamental theorems hold
- Since the production set Y is a convex set,
- and the utility function is strictly quasi-concave.
- Hence, the set of the walrasian equilibria and the pareto set coincide
- And, the walrasian equilibrium being unique at E,
- it must also be the only Pareto Efficient allocation.

□